

Venkata Saratchandra Patnaik Markonda Patnaikuni

+1 6023018122 | smarkond@asu.edu | <https://www.linkedin.com/in/saratchandra-92015>

EDUCATION

Master of Science in Computer Science (CGPA: 3.81/4)

December 2025

Arizona State University, Tempe, AZ

Bachelor of Technology in Computer Science (CGPA: 9.11/10)

May 2022

Christ University School of Engineering, Bengaluru, India

TECHNICAL SKILLS

GPU & Parallel Computing: CUDA Programming, PyCUDA, Shared Memory Optimization, Loop Unrolling, Tiling, Grid-Block-Thread Architecture

Programming & Scripting Languages: Python, C, C++, R, Java, Kotlin, Go, JavaScript, TypeScript

Machine Learning & AI Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, Matplotlib, OpenAI API

Cloud & DevOps Tools: AWS(S3, EKS, EC2, Greengrass, Lambda, Load Balancer), Docker, Kubernetes, Linux, Shell Scripting

CI/CD, Monitoring & Dev Tools: GitHub, GitOps, ArgoCD, Jira, Jenkins, Grafana

PROJECTS

Parallel Implementation of Gaussian Filter on GPU (CUDA, PyCUDA, Image Processing)

- Implemented 2D Gaussian convolution on GPU using CUDA & PyCUDA, optimizing performance with thread grids, blocks, and shared memory.
- Applied techniques like loop unrolling, tiling, and separable filters, achieving up to 20x speedup over a CPU-based MATLAB implementation.
- Validated results using PSNR and SSIM metrics; analyzed computation time for various kernel/image sizes to quantify performance gain.

Underground Cable Fault Detection (IoT and Embedded Systems)

- Designed and simulated a fault detection system for underground cables using NodeMCU ESP8266 and voltage sensing, implementing real-time monitoring with LCD and remote notifications.
- Applied embedded programming and control logic, simulating industrial automation workflows and system diagnostics.
- Integrated automation principles, including fault detection, alerting, and real-time response, demonstrating transferable PLC/HMI/SCADA skills.

Real-Time Face Recognition on AWS Greengrass (Cloud Computing Project)

- Built an edge-based face recognition system using AWS IoT Greengrass, MQTT, Lambda, and SQS, simulating smart camera-to-core communication over EC2.
- Deployed MTCNN and FaceNet models under pip-free constraints by integrating raw PyTorch source, optimizing runtime within AWS Greengrass component limits.
- Debugged a complex distributed pipeline using CLI tools and logs across Greengrass, MQTT, and Lambda. Automated deployments via shell scripts and passed over 100+ real-time autograder test cases.

Relevant Coursework

Computer Organization and Architecture, System Software, Operating Systems, Microprocessors & its Applications

PROFESSIONAL EXPERIENCE

Software Implementation Engineer: Amagi Media Labs, Bengaluru, India

August 2022 - November 2023

- Integrated StreamDeck client systems for real-time control of media dashboards, enabling operational efficiency in live broadcast workflows.
- Designed automation workflows using Terraform and Python scripts to streamline onboarding, reducing manual setup time by 90%.
- Managed 15+ containerized microservices on AWS EKS with Docker/Kubernetes; deployed CI/CD pipelines via GitOps and ArgoCD, improving reliability and reducing failures by 95%.
- Developed monitoring and diagnostic scripts for real-time systems, simulating industrial control and automation processes.